Loan Default Prediction

Capstone Project — Final Submission

Practical Data Science | MIT-PE Applied AI & Data Science Program, May 2026 cohort | Full-Code Track

1. Executive Summary

Most Important Findings

- The dataset (5,960 HELOC applicants) has a historical default rate of 19.95% (1,189 defaults). No data quality issues (duplicates) were found, but 11 of 12 predictors have missing values, most severely DEBTINC at 21.3%.
- The single most important finding in the entire analysis: applicants whose debt-to-income ratio (DEBTINC) could not be established defaulted at 62.0%, versus 8.6% for applicants who reported it — missingness itself is one of the strongest predictors of default in this dataset.
- Across four required model families, credit history variables — DELINQ (delinquent credit lines), DEROG (derogatory reports), DEBTINC, and CLAGE (credit history length) — consistently rank as the top default-risk drivers, confirmed across three independent explainability methods.
- Tuned XGBoost achieved the best balance of Recall (0.798), Precision (0.764), F1 (0.781), and ROC-AUC (0.950) on a held-out 30% test set — evaluated using a leakage-safe pipeline in which every preprocessing statistic is refit independently within each cross-validation fold — outperforming Logistic Regression, Decision Tree, and Random Forest on the metrics that matter most for this business problem.

Final Proposed Model Specification

XGBoost classifier, tuned via 5-fold cross-validated RandomizedSearchCV (25 iterations) optimizing Recall, with all preprocessing wrapped in a scikit-learn Pipeline so each fold refits its own imputation/scaling statistics. Final hyperparameters: n_estimators=300, max_depth=4, learning_rate=0.05, subsample=0.9, colsample_bytree=0.9, scale_pos_weight≈4.01. Trained on 27 features (10 numeric predictors, 11 missing-value indicator flags, 6 one-hot encoded job/reason categories) derived from the 12 original applicant fields. Recommended operating threshold range: 0.30–0.40 (versus the default 0.50), selected using cross-validated out-of-fold training probabilities and confirmed once on the held-out test set, to prioritize catching defaulters, pending validation against the bank's actual cost figures. Deployed alongside a Logistic Regression (class-weight balanced) companion model to support a model designed for explainable, defensible credit decisioning — subject to formal fair-lending and compliance validation before any production use.

Model	Test Recall	Test Precision	Test F1	Test ROC-AUC
Logistic Regression (Tuned)	0.742	0.545	0.629	0.868
Decision Tree (Tuned)	0.804	0.554	0.656	0.886
Random Forest (Tuned)	0.770	0.668	0.715	0.933
XGBoost (Tuned) — RECOMMENDED	0.798	0.764	0.781	0.950

2. Problem and Solution Summary

Summary of the Problem

The bank's consumer credit department currently relies on manual underwriting review to decide Home Equity Line of Credit (HELOC) applications — a process that is slow, inconsistent across underwriters, and vulnerable to human bias. The bank needs an empirically derived model, designed to support defensible and explainable credit

decisioning, that predicts which applicants are likely to default, so lending decisions can be made faster, more consistently, and with a data-grounded basis for any rejection — subject to the bank's own formal fair-lending and regulatory validation before that basis is used in an actual adverse-action decision.

Reasons the Proposed Solution Was Selected

XGBoost was not chosen simply because it produced the highest ROC-AUC — Random Forest's ROC-AUC (0.933) was close behind. XGBoost was selected because, once tuned toward Recall, it simultaneously delivered the best Recall (0.798), Precision (0.764), and F1 (0.781) of any tuned model, while its feature-importance output could be cross-validated against a fully transparent Logistic Regression model's coefficients — giving the bank both strong predictive performance and a credible path to interpretability, which a single "best metric" choice would not have guaranteed. Its output should be read as a risk-ranking score: the model discriminates well between likely and unlikely defaulters (ROC-AUC = 0.950), but its raw probabilities have not been verified as calibrated (a predicted 0.36 does not reliably mean a 36% observed default rate — see Limitations) and should not be quoted to stakeholders as literal probabilities of default.

How It Would Affect the Problem / Business

Deployed as a decision-support score — not a fully automated approve/decline switch — this model lets underwriters prioritize manual review time on the applications most likely to default, standardizes the criteria used across underwriters (reducing inconsistency), and creates an auditable, data-grounded basis intended to support the kind of specific, defensible reasoning ECOA adverse-action notices require — subject to the bank's own formal legal and compliance review before that basis is relied on in an actual notice.

3. Recommendations for Implementation

Key Recommendations

- Deploy the tuned XGBoost model as a decision-support score feeding into, not replacing, human underwriting judgment.
- Operate the model at a threshold in the 0.30–0.40 range rather than the 0.50 default, once Credit Risk validates the trade-off against real cost figures.
- Pair every model-driven decline with the Logistic Regression / feature-importance explanation, to support a defensible adverse-action rationale — pending formal compliance sign-off that the explanation meets ECOA's specific-reason requirement.
- Launch an operational investigation into why DEBTINC is missing for over a fifth of applicants, given how strongly that missingness alone predicts default — a process fix with standalone value.

Key Action Items by Stakeholder

- Credit Risk team: validate the proposed 0.30–0.40 threshold range against actual loss-given-default and cost-of-review figures; formally sign off on an operating threshold.
- Underwriting Operations: pilot the model in shadow mode (running alongside, not yet influencing, human decisions) before any live use.
- Compliance / Fair Lending: conduct a full disparate-impact review using demographic data this project did not have access to, before production deployment.
- Data / IT: investigate and remediate the DEBTINC data-collection gap.

Expected Benefit and Costs

This dataset does not contain loan interest rates, dollar amounts of realized losses, or the bank's cost of manual review — so a reliable dollar benefit estimate cannot be produced from this data alone, and none is claimed here. The framework below lets the Credit Risk team populate their own figures:

Expected benefit ≈ (additional true defaulters caught × average loss-given-default per default) − (additional false positives × average foregone profit per wrongly-declined applicant) − (cost of manual review time for all flagged applications)

For illustration only (from the held-out test set of 1,788 applicants, 357 actual defaulters): moving the final tuned XGBoost model's threshold from 0.50 to 0.30 changes outcomes from 285 true positives / 72 false negatives / 88 false positives to 317 true positives / 40 false negatives / 169 false positives — 32 more defaulters caught at the cost of 81 more false alarms (applications flagged for manual review that would not have defaulted). Whether this trade is worth it in dollars depends entirely on the bank's own per-unit cost figures, which this dataset does not contain.

Risks, Challenges, and Limitations

- The model was trained only on historically approved applicants — its behavior on applicants who would have been declined under the old process is untested (a selection-bias limitation inherent to this data).
- Median/mode imputation and IQR outlier capping are defensible but not the only valid preprocessing choice, and compress the tails of skewed variables somewhat.
- XGBoost is less intrinsically transparent than Logistic Regression; post-hoc feature importance is standard practice but not identical to true model transparency.
- No demographic fields exist in hmeq.csv — no fairness claim beyond REASON/JOB can be supported by this dataset; a full fair-lending review requires data this project does not have. This project does not claim, and cannot claim, that the model is free of disparate impact.
- Predicted probabilities have not been verified as calibrated: a Brier-score/calibration-curve check found strong discrimination (ROC-AUC 0.950) but over-stated risk in the mid-probability range (e.g., a mean predicted 0.36 vs. an observed default rate of 0.25). The model should be treated as a ranking/flagging tool, not a source of literal default probabilities, until a recalibration step is added.

Further Analysis Needed

- A cost-benefit analysis using the bank's actual financial figures.
- A fair-lending / disparate-impact review using demographic data not present in this dataset.
- A model monitoring and periodic retraining plan, since applicant population and macroeconomic conditions drift over time.
- A root-cause investigation into the DEBTINC missing-data pattern.
- A recalibration step (e.g., Platt scaling or isotonic regression) if the raw probability score, not just the resulting flag, is shown to underwriters.